9A Lesson 7

# Natural selection and evolution

Year 9 Science · 9A Genetics and evolution · Student Book pages 18 to 20

Name _____ Class _____ Date _____

**Today I can:**

- ☐ I can explain how natural selection works on inherited variation when the environment changes, using the peppered moth and the Salton Sea tilapia.
- ☐ I can describe the theory of evolution by natural selection and explain how natural selection can produce a new species.

**Key words**

**natural selection** organisms better adapted to their environment are more likely to survive, reproduce and pass on their genes

**evolution** a change over time in the characteristics of organisms, sometimes making a new species

**inherited variation** differences between individuals caused by genes, passed on to offspring

**gene** a section of DNA that carries the instructions for one characteristic

**reproduce** to produce offspring

**generation** all the offspring born at about the same time; parents, children and grandchildren are three generations

**hypothesis** an idea, based on observations, that can be tested

**theory** an explanation that has been tested many times and is accepted by scientists

**species** a group of organisms that can breed together to produce offspring that can also breed

**camouflage** colours or patterns that make an organism hard to see

**cloning** making a new organism with exactly the same genetic information as another

## ● Core Everyone does this section.

**1.** State what causes inherited variation.

**2.** Many times a year, a certain river suddenly floods. Over many years, a species of lizard living by the river evolved longer legs. Darwin might have explained this using the sentences in the table. Number them 1 to 5 in the correct order.

| Order | Sentence |
|---|---|
| | In the next generation, more of the lizards have longer legs. |
| | The surviving lizards produce offspring that also have longer legs. |
| | The lizards with slightly longer legs are more likely to survive the flood. |
| | This process of natural selection repeats over and over again. |
| | When the river floods, the lizards with slightly longer legs run faster. |

**3.** In the 1850s most peppered moths around Manchester were pale. By 1895, 98% were black. Factories built in those years covered the trees in soot.

**a)** Explain why black moths survived better than pale moths around Manchester at the end of the 19th century.

**b)** Manchester now has many fewer factories and the trees are no longer black. Predict how the number of black moths has changed, and use natural selection to explain your prediction.

## ●● Stretch When Core is finished.

**S1.** Lamarck and Darwin both tried to explain how wading birds evolved long legs. Lamarck said: 'Birds that could wade into deeper water got more fish. When there were fewer fish, the birds stretched their legs a bit, and passed their longer legs on to their offspring. Over time the birds' legs got longer and longer.' Lamarck's idea is not accepted. Write what Darwin would say instead. Use the words genes, variation, survive, reproduce and generation.

**S2.** The Salton Sea in California became saltier and saltier after it formed in 1905. Tilapia are the only fish still living there.

**a)** State the characteristic that has been naturally selected in the Salton Sea tilapia.

**b)** Explain how this selection happened.

## ●●● Challenge Explain in full sentences.

**C1.** Explain how natural selection can produce a new species if two populations of the same organism become separated, for example on two islands. (Hint: think about the definition of a species.)

**C2.** Woolly mammoths became extinct about 4000 years ago. Mammoths and elephants evolved from one species about 6 million years ago. Mammoths had 58 chromosomes.

**a)** How many chromosomes would a mammoth sperm cell have?

**b)** Suggest how mammoths may have evolved thick hair, when the species they evolved from had very little hair.

**c)** Explain why scientists could clone a mammoth from frozen remains but could not clone a dinosaur.

## My thinking (any language)

Someone says: 'Evolution is only a theory.' What does the word theory mean in science? Is 'only a theory' a good reason to doubt it?

**Word bank** natural selection · evolution · inherited variation · gene · reproduce · generation · hypothesis · theory · species · camouflage

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**Exit ticket 9A Lesson 7: Natural selection and evolution** Name _____

**1.** What is natural selection?

**2.** What is evolution?

**3.** Explain why dark peppered moths became more common near Manchester in the 1800s.

**Today I can:** Goal 1 ☐ Goal 2 ☐